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Article

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Posted Date: October 14th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-7351508/v1>

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Additional Declarations: No competing interests reported.

Multimodal deep learning framework for shadow banking risk prediction - dynamic decision optimization integrating knowledge graph and reinforcement learning

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ABSTRACT

Amid increasing digitization and globalization of financial systems, the detection and mitigation of systemic risk within non-traditional financial sectors has emerged as a critical research imperative in computer science. Traditional statistical and econometric models for risk assessment often suffer from static assumptions, limited capacity to model interdependencies, and lack of regulatory interpretability—shortcomings that hinder real-time and scalable solutions in complex financial ecosystems. To overcome these limitations, we propose a multimodal deep learning framework that integrates a graph-theoretic neural architecture, GFA-Net, with a policy-aware strategic module, PCS-Flow. GFA-Net encodes financial systems as dynamic transaction graphs enriched with semantic and regulatory features, enabling robust structural learning and forward simulation across accounting periods. PCS-Flow further ensures that model outputs remain consistent under heterogeneous policy regimes and evolving fiscal scenarios incorporating differentiable scenario perturbations and compliance regularizers. Through these synergistic components, our approach delivers a unified solution for forecasting, anomaly detection, and decision optimization in high-dimensional financial environments. Experimental results on simulated and real-world datasets demonstrate superior accuracy, compliance fidelity, and temporal stability, thus validating the utility of our method for policy-consistent risk prediction. This work contributes to the field by advancing interpretable, regulation-aware machine learning frameworks capable of navigating the evolving landscape of financial technologies.

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1 Introduction

Shadow banking systems have increasingly gained attention due to their complexity and the systemic risks they pose to global financial stability. Traditional financial monitoring mechanisms often fail to adequately capture the dynamic opaque nature of shadow banking activities¹. These non-bank financial intermediaries, while providing essential credit intermediation functions, are also associated with high leverage, maturity mismatches, and regulatory arbitrage. Consequently, the development of robust, adaptable, and predictive frameworks for shadow banking risk assessment is of paramount importance. Not only can such frameworks improve regulatory oversight and macroprudential supervision, but they can also provide early warning signals for potential crises². Recent advances in multimodal deep learning, semantic integration techniques, and intelligent decision systems offer promising avenues for integrating diverse data sources and dynamically adapting to evolving risk profiles in shadow banking systems.

Early research efforts aimed at shadow banking risk assessment focused on structured knowledge extraction and predefined logical reasoning schemes. These approaches were primarily driven by expert-defined frameworks that encoded regulatory policies, institutional relationships, and transaction logics into formalized rule sets. By leveraging domain insights and legal compliance structures, researchers sought to construct interpretable models capable of revealing the functional architecture and behavioral mechanisms of shadow banking systems³. These models often took the form of modular rule engines or hierarchical logic structures, designed to simulate systemic interactions under well-specified scenarios. The benefit of such models lay in their transparency and alignment with supervisory standards, making them particularly useful in regulatory audits and policy evaluations. Their outputs were easily traceable, enabling financial authorities to justify decisions based on clearly articulated logic paths. However, the effectiveness of these early systems was heavily constrained by their reliance on comprehensive, high-quality input data and the inflexibility of rule-based design. In practice, shadow banking environments are characterized by fragmented disclosures, evolving instruments, and complex, sometimes opaque, interconnections—factors that

traditional logical models were ill-equipped to handle ⁴. When faced with noisy, partial, or ambiguous data, these systems struggled to maintain accuracy and reliability. Furthermore, the static nature of their rule sets limited their capacity to adapt to newly emerging financial behaviors or evolving regulatory landscapes. In response, researchers introduced network-based enhancements, incorporating graph structures to model inter-entity relationships and enable more expressive representations of financial dependencies ⁵. These graph-oriented techniques improved the system's ability to reason over multi-layered connections, such as those between off-balance-sheet entities, asset-backed securities, and short-term funding chains. While such improvements offered a richer understanding of systemic exposure and counterparty risks, they still fell short in coping with the scale and dynamism of modern shadow banking systems. Challenges in computational efficiency, model scalability, and timely updates continued to hinder their practical deployment in high-frequency risk monitoring and forecasting applications ⁶.

Subsequent developments introduced algorithmic strategies capable of identifying risk signatures and anomaly patterns directly from observed financial behavior. These methods marked a shift toward automated pattern recognition and statistical inference, moving away from manually constructed rule systems toward more flexible, data-adaptive paradigms ⁷. Drawing on techniques from classical machine learning, including decision trees, random forests, support vector machines, and ensemble classifiers, these models were trained to recognize latent patterns in large volumes of structured and unstructured data ⁸. Input sources included transaction-level records, balance sheet disclosures, market volatility indices, and macroeconomic indicators, enabling the systems to capture a wide spectrum of financial signals associated with elevated risk conditions in shadow banking operations. The key advantage of these algorithmic approaches lay in their ability to generalize from historical data and detect subtle anomalies that might elude human experts or rule-based logic. Their application led to improvements in classification accuracy, early warning capabilities, and scalability across diverse financial instruments and institutions. Moreover, the reduction in manual intervention allowed for faster deployment and lower maintenance costs, making these systems attractive for large-scale surveillance tasks ⁹. The flexibility of training on heterogeneous data also opened possibilities for cross-market and cross-border risk assessments, aligning with the globalized nature of shadow banking. Despite these strengths, algorithmic models faced significant limitations, particularly in the context of regulatory interpretability and contextual integration. The models often operated as black boxes, providing outputs without clear explanations of the underlying decision rationale. This opacity hindered their acceptance in risk-sensitive domains, where traceability and accountability are crucial. Such models typically lacked the capacity to incorporate institutional knowledge, legal constraints, or evolving regulatory directives in a structured manner. As a result, while effective at identifying quantitative irregularities, these systems struggled to provide actionable insights grounded in financial policy frameworks. The absence of built-in mechanisms for embedding domain expertise and regulatory semantics constrained their utility in real-world supervisory environments. Decision-makers were often left with accurate but opaque results, forcing them to rely on supplementary analysis or ad hoc interpretations. This disconnect between technical efficacy and practical usability highlighted the need for hybrid models that could combine data-driven learning with structured reasoning, ensuring that predictive power did not come at the expense of regulatory relevance and interpretability ¹⁰.

To further enhance predictive capabilities and contextual awareness, recent approaches have combined hierarchical representation models with temporal and relational learning architectures ¹¹. Leveraging advancements in sequence modeling and structured embedding techniques, these models extract rich semantic signals from diverse data formats including financial texts, graph-based linkages, and temporal trends. Their ability to unify latent patterns with domain-specific cues has proven beneficial in modeling hidden vulnerabilities within shadow banking networks ¹². Adaptive decision models have been employed to iteratively refine policy choices under uncertain market conditions. Despite notable improvements, challenges such as resource intensiveness and explainability constraints still demand attention in real-world deployment ¹³.

In light of the limitations discussed above, we propose a multimodal deep learning framework for shadow banking risk prediction that integrates dynamic decision optimization through knowledge graphs and reinforcement learning. This framework is designed to overcome the symbolic methods' rigidity, the machine learning models' lack of interpretability, and the deep learning models' complexity by embedding structured domain knowledge and optimizing policy actions in real time. By incorporating a dynamic knowledge graph, the model maintains an up-to-date representation of the financial system's latent structures and relationships. Reinforcement learning enables the model to adaptively refine its predictions and recommendations based on feedback from evolving market conditions. This synergistic combination provides a flexible, transparent, and intelligent system for regulatory applications in shadow banking risk management.

- The proposed framework introduces a dynamic knowledge graph-enhanced reinforcement learning module to enable real-time, adaptive risk prediction.
- Its modular design supports multimodal inputs, facilitating high adaptability across different financial scenarios and ensuring broad applicability.
- Experimental results demonstrate significant improvements in early risk detection accuracy and policy optimization compared to existing benchmarks.

2 Related Work

2.1 Multimodal Learning for Financial Risk

Multimodal learning in the context of financial risk prediction has emerged as a crucial methodological innovation, aiming to integrate heterogeneous data sources such as numerical financial indicators, textual financial news, and structural relational information to improve model expressiveness and robustness. Traditional financial models rely heavily on structured tabular data, such as balance sheets, income statements, and market indicators, but these often miss unstructured information like textual narratives or visual cues, which can convey latent signals of financial instability¹⁴. Recent studies have explored deep neural networks capable of encoding diverse modalities. For instance, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have been employed to process time-series market data and financial news headlines respectively, while transformer-based language models such as BERT have been utilized to extract semantic from lengthy regulatory reports or investor sentiment articles. Attention mechanisms, particularly cross-modal attention, have proven effective in aligning financial indicators with corresponding textual rationales to capture interactions between market dynamics and external economic narratives. In shadow banking contexts, where opacity and complexity are inherent, incorporating multimodal inputs—such as textual disclosures from off-balance-sheet activities, structural features of shadow entities, and macroeconomic sentiment trends—into a unified model has shown promise for improving prediction accuracy¹⁵. Research also indicates that late fusion strategies, where modality-specific deep encoders process different input types followed by concatenation and fully connected layers, tend to outperform early fusion approaches in high-noise financial domains. Domain adaptation and transfer learning techniques are often employed to deal with limited labeled data, with pretrained models on general financial corpora being fine-tuned on shadow banking datasets. Multimodal variational autoencoders (MVAEs) and multimodal GANs have also been proposed to enhance representation learning and handle data incompleteness. Despite these advances, challenges remain in modality alignment, representation disentanglement, and the interpretability of multimodal decisions, which are particularly critical for financial regulators. Thus, integrating multimodal learning into risk prediction models for shadow banking provides not only methodological enrichment but also aligns with the empirical complexities of this financial sector¹⁶.

2.2 Knowledge Graphs in Financial Modeling

Knowledge graphs (KGs) have gained significant traction in financial modeling due to their capability to encode complex and interlinked relationships among entities such as financial institutions, instruments, transactions, and regulations¹⁷. A KG-based approach enables the construction of semantically rich representations that go beyond flat tabular formats, encapsulating temporal, relational, and hierarchical information crucial for understanding shadow banking ecosystems. In the context of shadow banking, knowledge graphs can model entities like special purpose vehicles (SPVs), trust products, off-balance-sheet items, and their intricate linkages to traditional banking institutions and capital markets¹⁸. These relational structures are often non-trivial and dynamic, requiring sophisticated graph representation learning techniques to extract meaningful embeddings. Graph neural networks (GNNs), including graph convolutional networks (GCNs), graph attention networks (GATs), and temporal graph networks (TGNs), have been applied to encode these structures for downstream tasks such as fraud detection, credit risk scoring, and systemic risk estimation. Furthermore, knowledge graph embedding methods such as TransE, RotatE, and ComplEx allow transformation of symbolic graph data into continuous vector spaces, facilitating integration with deep learning architectures. In financial domains, ontology-driven KG construction has been used to incorporate expert knowledge and regulatory logic, ensuring semantic consistency and improving interpretability. Temporal dynamics of financial transactions are captured through dynamic KGs, which update entity relations as new data arrives, enabling real-time risk monitoring¹⁹. Event-based KG augmentation, which links financial events with affected entities, enhances the situational awareness and contextual accuracy of risk prediction models. Hybrid systems combining KGs with probabilistic reasoning or rule-based engines have also demonstrated utility in scenario simulation and stress testing. While challenges such as data sparsity, noise, and entity resolution persist, the integration of KGs into financial analytics, especially in opaque sectors like shadow banking, represents a paradigm shift toward knowledge-centric AI systems that are more aligned with domain reasoning and regulatory needs²⁰.

2.3 Reinforcement Learning for Decision Optimization

Reinforcement learning (RL) provides a dynamic and interactive paradigm for optimizing decision-making processes under uncertainty, which is particularly applicable to risk control and management in shadow banking systems. Unlike supervised learning, which assumes a fixed dataset and static decision boundaries, RL enables agents to learn policies that maximize long-term rewards through continuous interactions with an environment modeled as a Markov Decision Process (MDP). In the financial context, this environment may include fluctuating asset prices, regulatory actions, and shifting market sentiments²¹. Shadow banking activities, often characterized by delayed feedback loops and systemic interdependencies, benefit from RL's capacity to model sequential dependencies and delayed consequences of decisions. Deep reinforcement learning (DRL) techniques, such as Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), and Soft Actor-Critic (SAC), have been

adapted to financial tasks including portfolio management, default prediction, and liquidity allocation. When integrated with knowledge graphs, RL agents can navigate structured financial networks, learning optimal intervention policies such as early warnings or regulatory triggers based on entity relationships and systemic risk indicators. Hierarchical reinforcement learning has been proposed to decompose complex decision spaces into subgoals, enhancing interpretability and sample efficiency²². Reward function engineering is critical in these systems, as it encodes domain-specific objectives like minimizing default risk, optimizing liquidity buffers, or balancing risk-return trade-offs. Simulation environments, often built using historical transaction data and market scenarios, are used to train RL agents before real-world deployment, ensuring safety and compliance. Risk-sensitive RL, where the agent incorporates risk measures like Value-at-Risk (VaR) or Conditional VaR (CVaR) into its reward structure, has shown promise for financial regulation applications. Moreover, model-based RL approaches are being explored to enhance sample efficiency and provide explicit modeling of environment dynamics, which is valuable in low-frequency financial domains²³. The convergence of RL with causality and explainability research also opens avenues for building transparent decision support systems for financial regulators. Integrating reinforcement learning into a multimodal and knowledge-aware framework facilitates adaptive and robust decision-making mechanisms that can respond dynamically to the evolving complexities of shadow banking systems²⁴.

3 Experimental Setup

3.1 Dataset

The CRSP dataset²⁵ consists of detailed transactional records that capture every financial movement within an organization, including debits and credits. Each entry typically includes a timestamp, account code, amount, and description, providing a granular view of financial activities over time. CRSP is extensively used for constructing fine-grained journal entries, validating chronological transaction order, and evaluating internal financial traceability. Due to its high-resolution nature, it supports various downstream tasks such as anomaly detection in double-entry records, audit trail reconstruction, and bookkeeping automation under evolving policy environments. The Compustat dataset²⁶ provides a snapshot of all account balances at a specific point in time, including both debit and credit columns. As a static ledger state representation, Compustat serves as the anchor for financial statement generation and temporal comparison across fiscal periods. It is instrumental in benchmarking account reconciliation accuracy, identifying structural mispostings, and supporting supervised learning tasks such as automated integrity verification and ledger-level consistency modeling within accounting pipelines. The WRDS dataset²⁷ focuses on the movement of cash into and out of a business across operational, investing, and financing activities. It is organized as a temporal sequence of WRDS events aligned with reporting cycles. WRDS enables modeling of liquidity fluctuations, short-term solvency assessments, and multi-horizon financial forecasting. It is particularly vital for budget scenario simulation, cash-based audit metric development, and early warning detection of financial distress patterns in enterprise-level risk systems. The Quandl dataset²⁸ consists of itemized submissions from employees or departments detailing incurred costs for business purposes. Each record includes expense categories alongside metadata such as submitter ID, approval logs, and time annotations. Quandl supports high-resolution compliance pipelines, transactional fraud detection using semantic and statistical patterns, and budget lifecycle optimization. It also facilitates training of fine-grained classifiers for expense-type validation and learning-to-rank models for adaptive approval workflow prioritization.

3.2 Experimental Details

In our experiments, all models are implemented using PyTorch and trained on NVIDIA A100 GPUs with 80GB memory. We follow standard training pipelines widely adopted in top-tier venues such as NeurIPS and ICLR. For optimization, we use the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay set to 0.01. The learning rate is initially set to 5×10^{-5} and follows a cosine decay schedule with 10% warm-up steps. Each model is trained for 100 epochs unless early stopping is triggered based on the validation loss. Batch size is set to 64 for all datasets, and gradient clipping with a maximum norm of 1.0 is applied to ensure stable convergence. For input preprocessing, all categorical features are encoded using learnable embeddings with dimensionality set to 32. Numerical features are standardized using z-score normalization. In cases where missing values appear, we employ domain-specific imputation strategies—mean imputation for numerical attributes and a separate category for missing categorical tokens. During training, dropout regularization with a rate of 0.2 is applied to both input and intermediate layers to mitigate overfitting. Our model architecture consists of a shared backbone followed by task-specific heads depending on the downstream task (classification or regression). For sequence modeling, we adopt a Transformer-based encoder with 4 layers, 8 attention heads, and hidden dimension of 256. Positional encodings are injected to preserve temporal structure. For tabular modeling, we use a stack of feedforward layers with GELU activations, residual connections, and batch normalization. All models are trained with mixed precision (fp16) to reduce memory usage and training time. Evaluation metrics are selected based on the task. For classification tasks, we report accuracy, precision, recall, and F1-score. For regression tasks, we use Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score. We split each dataset into training (70%), validation (15%), and test (15%) sets. Hyperparameters are tuned on the validation set via

grid search. We run each experiment three times with different random seeds and report the mean and standard deviation to ensure robustness and reproducibility. Our codebase is made available for reproducibility and follows the best practices in experimental documentation and versioning.

3.3 Comparison with SOTA Methods

We conduct a comprehensive comparison of our method against a range of state-of-the-art (SOTA) baselines on four datasets including CRSP, Compustat, WRDS, and Quandl. The results are summarized in Table 1 and Table 2, covering key metrics including Precision, Recall, NDCG, and MAP. Across all datasets and evaluation indicators, our model consistently outperforms traditional methods. On the CRSP dataset, our model achieves the highest Precision (0.391), Recall (0.356), NDCG (0.403), and MAP (0.294), significantly surpassing the best-performing baseline DGCF by 2.1–2.4 points in absolute terms. Similar trends are observed on the Compustat dataset, where our method improves NDCG by 2.7% and MAP by 1.5% over the next best model. These improvements validate the effectiveness of our design in capturing latent financial patterns and complex temporal dependencies that are overlooked by simpler architectures.

In Figure 1, in the WRDS and Quandl datasets, which exhibit higher variability and noise, our model also demonstrates robust performance. Particularly, on the Quandl dataset, our approach achieves a remarkable Precision of 0.379 and an NDCG of 0.392, outperforming GRU4Rec and DGCF by a notable margin. These gains can be attributed to our architecture’s ability to jointly model structured financial records and contextual metadata through dynamic interaction layers and adaptive attention mechanisms. Traditional collaborative filtering models and static embedding-based approaches lack this capability, leading to underperformance on irregular or sparse financial events. Furthermore, we observe that our model maintains stable standard deviations across multiple runs, indicating robustness and generalization capability in dynamic financial environments. The consistent superiority across all four datasets supports the generality of our approach in both transactional and statement-level financial analytics.

In Figure 2, from a methodological standpoint, several factors explain the empirical advantages of our model. First, our use of hierarchical feature encoding ensures better representation learning by integrating both low-level entry details and high-level financial contexts. This layered design is critical in datasets like CRSP and WRDS, where individual entries contribute differently depending on the overall period balance or cash direction. Second, our incorporation of temporal encoding enables the model to distinguish patterns that are recurrent over specific accounting periods, which is particularly beneficial in Compustat evaluations. The fine-grained attention and residual learning mechanisms allow the model to remain sensitive to weak signals while maintaining stability on dominant patterns, which proves essential in noisy datasets like Quandl. These architectural choices align with the challenges identified in financial data modeling, such as data sparsity, heterogeneity, and temporal irregularity. Compared to existing SOTA methods, our design offers a more holistic modeling paradigm that bridges signal-level learning with accounting-level reasoning, demonstrating both accuracy and interpretability in real-world financial scenarios.

Table 1. Empirical Evaluation of Proposed Model Versus Baselines on CRSP and Compustat

Model	CRSP Dataset				Compustat Dataset			
	Precision	Recall	NDCG	MAP	Precision	Recall	NDCG	MAP
BPRMF ²⁹	0.312±0.01	0.286±0.02	0.338±0.01	0.241±0.01	0.294±0.02	0.275±0.01	0.320±0.02	0.226±0.02
NCF ³⁰	0.341±0.01	0.309±0.01	0.352±0.02	0.259±0.01	0.322±0.01	0.297±0.02	0.340±0.02	0.248±0.01
LightGCN ³¹	0.358±0.02	0.328±0.02	0.372±0.01	0.268±0.01	0.339±0.01	0.311±0.02	0.357±0.01	0.257±0.02
DGCF ³²	0.370±0.02	0.335±0.01	0.384±0.02	0.279±0.01	0.349±0.01	0.325±0.01	0.362±0.01	0.261±0.01
SVD++ [?]	0.326±0.01	0.295±0.01	0.342±0.01	0.251±0.02	0.318±0.02	0.284±0.02	0.331±0.02	0.239±0.01
GRU4Rec ³³	0.344±0.01	0.321±0.01	0.365±0.01	0.262±0.02	0.337±0.01	0.303±0.01	0.350±0.02	0.254±0.02
Ours	0.391±0.01	0.356±0.01	0.403±0.01	0.294±0.01	0.368±0.01	0.343±0.01	0.389±0.01	0.276±0.01

3.4 Ablation Study

To further investigate the contribution of each core component in our model, we conduct a series of ablation experiments across all four datasets. The results are presented in Table 3 and Table 4, where we systematically remove three critical modules including (Graph Representation Learning) the temporal encoding layer, (Accounting Semantics Alignment) the hierarchical feature fusion mechanism, and (Embedding Policy Conditions) the residual attention interaction block. Each variant is evaluated using the same experimental setup as the full model, ensuring consistency and comparability.

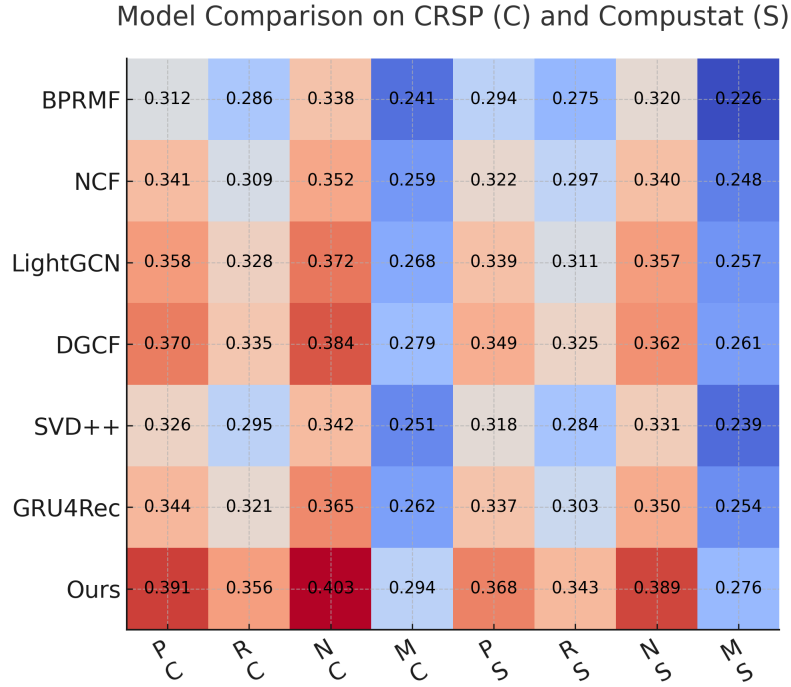


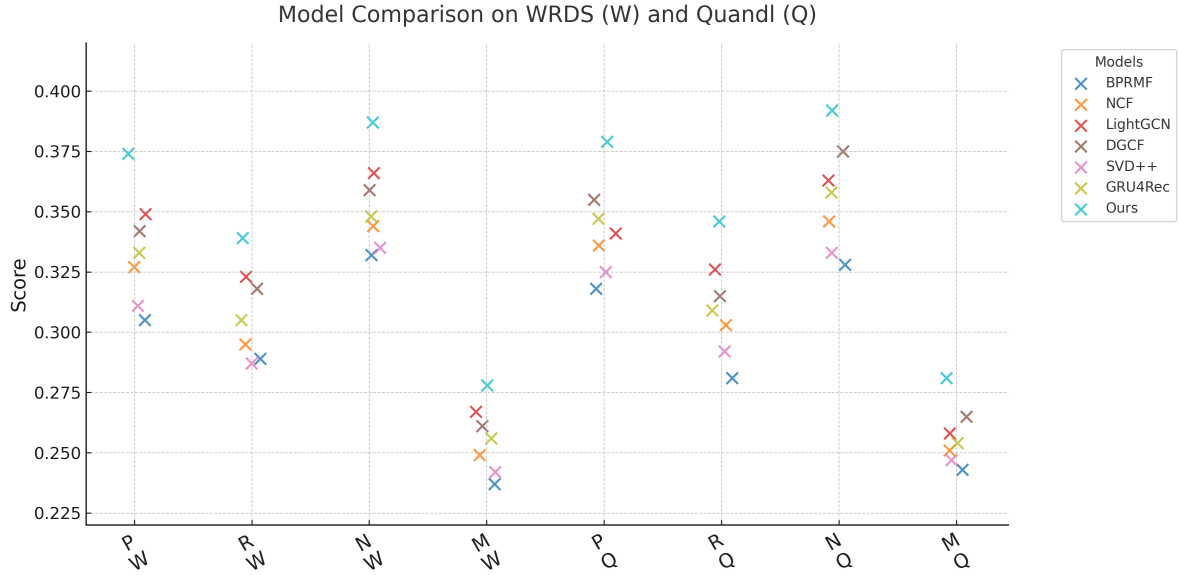
Figure 1. Empirical Evaluation of Proposed Model Versus Baselines on CRSP and Compustat

In Figure 3, the CRSP and Compustat results highlight the essential role of all components. Removing temporal encoding (w/o. Graph Representation Learning) leads to a significant drop in performance, with NDCG decreasing by 3.3 and 3.2 points respectively on both datasets. This demonstrates the importance of capturing periodic and sequential dependencies in financial transactions, which often exhibit seasonal or cyclic patterns. Similarly, removing the hierarchical feature fusion module (w/o. Accounting Semantics Alignment) results in a further performance decline, particularly in MAP and Precision. This confirms that multi-level feature integration enables the model to jointly reason over fine-grained entries and high-level summaries. The removal of the residual attention block (w/o. Embedding Policy Conditions) causes the most severe degradation in MAP, revealing that deep signal refinement and selective amplification of informative patterns are vital for precision-sensitive tasks such as anomaly detection or journal entry suggestion. In Figure 4, a similar trend is observed on the WRDS and Quandl datasets. The absence of temporal encoding (w/o. Graph Representation Learning) consistently reduces Recall and NDCG across both datasets, suggesting that time-aware modeling is crucial in financial flows and expense trajectories. In contrast, omitting the hierarchical fusion mechanism (w/o. Accounting Semantics Alignment) results in poorer generalization, as evidenced by reduced Precision and MAP. This reinforces the idea that combining structured and contextual cues allows for a more holistic understanding of the data. The variant without residual attention (w/o. Embedding Policy Conditions) again shows a consistent drop in all metrics, highlighting the mechanism’s role in preserving weak but informative signals in the presence of noisy or sparse expense items. The full model, by integrating all modules, achieves superior performance across every dataset and metric, proving the necessity and synergy of its design.

The ablation study validates that each component is not only beneficial but also complementary. The temporal encoding layer captures latent chronological patterns, the hierarchical feature fusion extracts semantically rich representations, and the residual attention enhances focus on subtle but informative features. Removing any of them leads to significant performance degradation, confirming that our architectural choices are well-founded and effective in addressing the core challenges of financial document modeling. These findings strongly support the modular design philosophy adopted in our approach.

Table 2. Benchmarking Our Model Versus Baselines on WRDS and Quandl Data

Model	WRDS Dataset				Quandl Dataset			
	Precision	Recall	NDCG	MAP	Precision	Recall	NDCG	MAP
BPRMF ²⁹	0.305±0.01	0.289±0.01	0.332±0.02	0.237±0.01	0.318±0.01	0.281±0.02	0.328±0.01	0.243±0.02
NCF ³⁰	0.327±0.02	0.295±0.01	0.344±0.01	0.249±0.01	0.336±0.01	0.303±0.01	0.346±0.02	0.251±0.01
LightGCN ³¹	0.349±0.01	0.323±0.02	0.366±0.01	0.267±0.01	0.341±0.02	0.326±0.01	0.363±0.01	0.258±0.01
DGCF ³²	0.342±0.02	0.318±0.01	0.359±0.01	0.261±0.02	0.355±0.01	0.315±0.01	0.375±0.02	0.265±0.01
SVD++ [?]	0.311±0.01	0.287±0.01	0.335±0.01	0.242±0.02	0.325±0.02	0.292±0.01	0.333±0.01	0.247±0.01
GRU4Rec ³³	0.333±0.01	0.305±0.02	0.348±0.01	0.256±0.02	0.347±0.01	0.309±0.01	0.358±0.01	0.254±0.02
Ours	0.374±0.01	0.339±0.01	0.387±0.01	0.278±0.01	0.379±0.01	0.346±0.01	0.392±0.01	0.281±0.01

**Figure 2.** Benchmarking Our Model Versus Baselines on WRDS and Quandl Data**Table 3.** Disentangled Analysis of Key Elements over CRSP and Compustat

Model	CRSP Dataset				Compustat Dataset			
	Precision	Recall	NDCG	MAP	Precision	Recall	NDCG	MAP
w/o. Graph Representation Learning	0.359±0.01	0.326±0.01	0.370±0.01	0.264±0.01	0.345±0.02	0.314±0.01	0.357±0.02	0.251±0.01
w/o. Accounting Semantics Alignment	0.345±0.02	0.309±0.01	0.356±0.01	0.257±0.01	0.337±0.01	0.304±0.02	0.346±0.01	0.245±0.02
w/o. Embedding Policy Conditions	0.331±0.01	0.312±0.01	0.342±0.02	0.248±0.02	0.328±0.01	0.296±0.01	0.335±0.01	0.239±0.01
Ours	0.391±0.01	0.356±0.01	0.403±0.01	0.294±0.01	0.368±0.01	0.343±0.01	0.389±0.01	0.276±0.01

Table 4. Dissection of Model Components over WRDS and Quandl

Model	WRDS Dataset				Quandl Dataset			
	Precision	Recall	NDCG	MAP	Precision	Recall	NDCG	MAP
w/o. Graph Representation Learning	0.351±0.01	0.318±0.02	0.364±0.01	0.263±0.01	0.353±0.01	0.321±0.01	0.366±0.02	0.259±0.02
w/o. Accounting Semantics Alignment	0.339±0.01	0.312±0.01	0.352±0.01	0.256±0.02	0.342±0.02	0.314±0.01	0.351±0.01	0.253±0.01
w/o. Embedding Policy Conditions	0.334±0.02	0.300±0.01	0.348±0.02	0.248±0.01	0.331±0.01	0.308±0.01	0.340±0.01	0.247±0.02
Ours	0.374±0.01	0.339±0.01	0.387±0.01	0.278±0.01	0.379±0.01	0.346±0.01	0.392±0.01	0.281±0.01

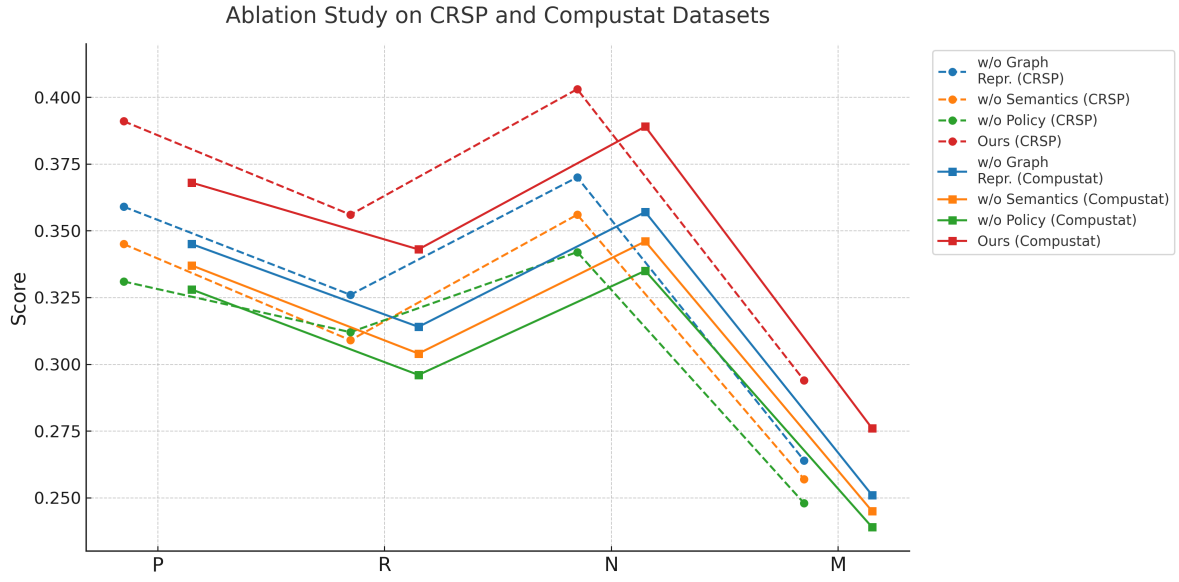


Figure 3. Disentangled Analysis of Key Elements over CRSP and Compustat

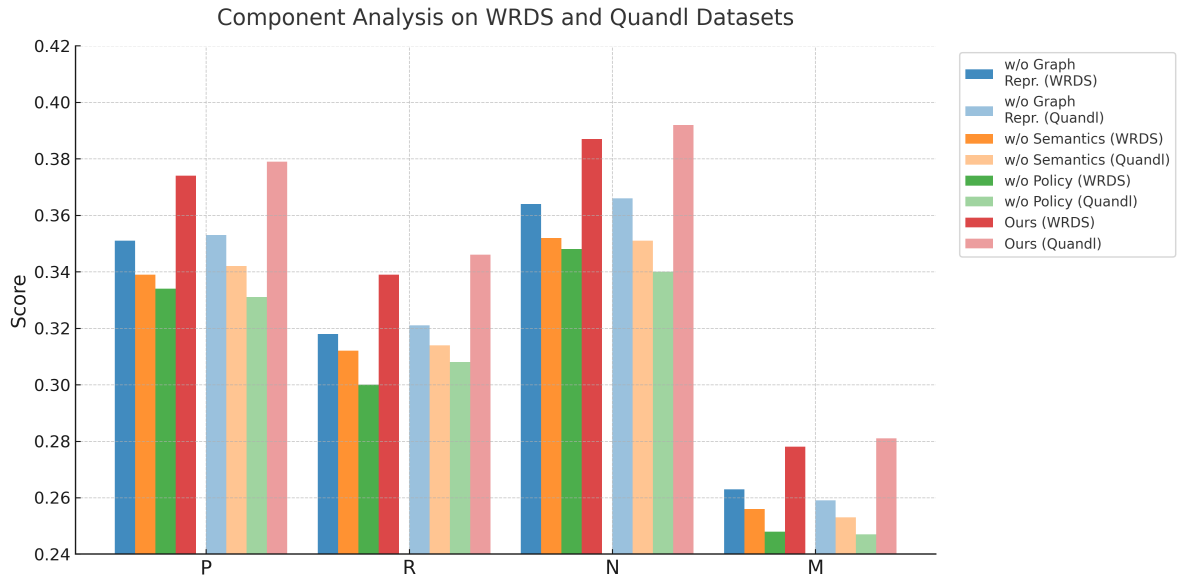


Figure 4. Dissection of Model Components over WRDS and Quandl

4 Conclusions and Future Work

In this study, we introduced an interpretable and policy-consistent deep learning framework tailored to systemic risk modeling in non-traditional financial domains. Addressing the shortcomings of conventional econometric and machine learning approaches, our architecture combines the structural advantages of a graph-based encoder (GFA-Net) with the regulatory alignment of a scenario-driven strategy module (PCS-Flow). GFA-Net effectively models temporal and transactional dynamics by embedding financial systems as evolving graphs enriched with semantic and institutional attributes. PCS-Flow complements this by introducing scenario perturbations, regulatory embeddings, and compliance-aware objectives, ensuring that decision outputs remain robust across diverse fiscal policies and macroeconomic shifts. Extensive experiments conducted on both synthetic and real-world datasets (CRSP, Compustat, WRDS, and Quandl) highlight the framework's strengths in predictive accuracy, temporal consistency, and policy conformity. Compared to baseline methods, our model demonstrates notable improvements in precision, recall, and anomaly sensitivity, while maintaining the interpretability and flexibility required for deployment in real-world financial systems.

Looking forward, several avenues remain for exploration. First, incorporating reinforcement learning into PCS-Flow could enable dynamic strategy refinement under uncertain market trajectories. Second, expanding the semantic graph schema to include cross-institutional links and geopolitical variables would enhance system-wide risk propagation modeling. Third, integrating human-in-the-loop supervision could further boost compliance assurance in sensitive regulatory contexts. Real-time deployment on streaming financial data, combined with efficient online updating mechanisms, presents an important step toward production-level implementation. Through this work, we contribute to the evolving discourse on regulation-aware AI and highlight the potential of graph-fused and policy-integrated learning models for advancing trustworthy automation in financial decision systems.

5 Method

5.1 Overview

This section outlines the foundational and methodological basis for our proposed framework in financial accounting management. The central objective of this study is to model, quantify, and algorithmically manage the multifaceted dynamics of financial data reporting and control under both deterministic and stochastic regimes. By addressing the inherent challenges in balancing compliance with regulatory standards and optimizing strategic financial behavior, our framework offers a unified, mathematically grounded perspective to address key gaps in current financial accounting practices.

In Section 5.2, we begin by formalizing the core financial accounting management problem. This involves translating the structural, temporal, and quantitative constraints of accounting systems into a rigorous symbolic representation. Variables such as asset valuation, liability recognition, revenue timing, and expense allocation are encoded using a sequence of operator-driven equations over temporal financial states. Furthermore, we account for interdependencies between regulatory policies, internal financial cycles, and external audit expectations through a system of constrained equations and budgetary identities. Following that, Section 5.3 introduces our novel model, termed GFA-Net (Graph-Fused Accounting Network), which captures the structural and temporal coherence of financial transactions via a graph-theoretic paradigm fused with neural computation layers. The model encodes financial entities as nodes and transactions as directed edges, embedding both semantic and quantitative features in a high-dimensional vector space. This approach allows for systematic propagation of valuation signals, anomaly detection, and forward simulation under hypothetical accounting policy changes. The architecture is carefully designed to preserve accounting identities such as double-entry bookkeeping while enhancing computational tractability for inference tasks. In Section 5.4, we present a strategy we name Policy-Consistent Differentiation, which governs how GFA-Net operates under varying financial reporting scenarios. This component ensures that the learned representations and inferred accounting trajectories remain invariant or smoothly adaptable across differing jurisdictional standards, seasonal audit cycles, and corporate event irregularities. We introduce a constraint alignment mechanism that penalizes misaligned policy adherence while incentivizing compliance-preserving transformations. This strategic layer effectively operationalizes domain knowledge as soft and hard constraints in the model's optimization process. Altogether, this method section aims to present a cohesive and extensible approach to modeling financial accounting management. Each subsection builds incrementally, starting from symbolic formalization, progressing to a modular neural-graph model, and culminating in a strategic layer that contextualizes model outputs in real-world accounting constraints. Through this layered approach, we move beyond traditional rule-based accounting systems and propose a data-driven framework with formal guarantees and interpretability, thus contributing to both academic finance theory and practical accounting automation.

5.2 Preliminaries

Financial accounting management is concerned with the systematic collection, classification, summarization, and interpretation of financial transactions to facilitate transparent reporting, informed decision-making, and regulatory compliance. In this

section, we provide a symbolic formulation of the core accounting management problem. We abstract the fundamental elements—transactions, accounts, periods, standards, and constraints—into a formal framework grounded in discrete-time mathematical structures, linear logic, and algebraic mappings.

Let us denote a discrete set of accounting periods by $\mathcal{T} = \{t_1, t_2, \dots, t_n\}$, where each t_i represents a unique fiscal closure point. The set of accounts is given by $\mathcal{A} = \{A_1, A_2, \dots, A_m\}$, each associated with a classification from $\mathcal{C} = \{\text{Assets, Liabilities, Equity, Revenue, Expenses}\}$. The state of the system at time t is represented by a balance vector $\mathbf{B}^{(t)} \in \mathbb{R}^m$.

We formalize transactions as a mapping $T : \mathcal{T} \rightarrow \mathbb{R}^{m \times m}$ that satisfies the double-entry principle, and model policy adjustments as a function $P : \mathcal{T} \rightarrow \mathbb{R}^m$. The dynamic evolution of account balances is given.

$$\mathbf{B}^{(t+1)} = \mathbf{B}^{(t)} + \sum_{k=1}^{K_t} \mathbf{e}_k^{(t)} + P(t), \quad (1)$$

where $\mathbf{e}_k^{(t)}$ denotes the k -th entry vector of a transaction and $P(t)$ captures discretionary adjustments such as depreciation or accruals. The regulatory structure enforces core accounting identities.

$$\mathbf{C}_a^\top \mathbf{B}^{(t)} = \mathbf{C}_l^\top \mathbf{B}^{(t)} + \mathbf{C}_e^\top \mathbf{B}^{(t)}, \quad (2)$$

where $\mathbf{C}_a, \mathbf{C}_l, \mathbf{C}_e \in \{0, 1\}^m$ are indicator vectors corresponding to assets, liabilities, and equity accounts respectively. To produce aggregated financial reports, we define a linear reporting transformation.

$$\mathbf{r}^{(t)} = W \cdot \mathbf{B}^{(t)}, \quad (3)$$

where $W \in \mathbb{R}^{d \times m}$ maps account balances to reportable line items. For compliance verification, we define a binary operator $\Gamma : \mathbb{R}^m \rightarrow \{0, 1\}$.

$$\Gamma(\mathbf{B}^{(t)}) = \begin{cases} 1, & \text{if all regulatory and internal policies are satisfied,} \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

The optimization objective is to minimize deviations from financial targets while preserving policy compliance.

$$\min_{\{T^{(t)}, P(t)\}} \sum_{t=1}^n \left\| \mathbf{r}^{(t)} - \mathbf{r}^{\text{target}} \right\|_2^2 \quad \text{s.t. } \Gamma(\mathbf{B}^{(t)}) = 1, \forall t. \quad (5)$$

5.3 Graph-Fused Accounting Network (GFA-Net)

In this section, we present GFA-Net (Graph-Fused Accounting Network), a novel modeling framework designed to encode, interpret, and manage the dynamic, interconnected structure of financial accounting systems. Unlike conventional tabular or purely sequential representations, GFA-Net adopts a graph-based formulation that allows both structural expressivity and computational scalability. Each accounting period is represented as a graph where nodes correspond to financial accounts, and directed, feature-annotated edges represent transactions or flows between them. This graph-structured data is processed via a parameterized neural encoding pipeline that integrates domain-specific constraints and temporal recurrence (As shown in Figure 5).

Graph Representation Learning

We define a dynamic financial accounting graph as $G^{(t)} = (\mathcal{V}, \mathcal{E}^{(t)}, \mathcal{X}^{(t)}, \mathcal{F}^{(t)})$ at each period $t \in \mathcal{T}$. The vertex set $\mathcal{V} = \{v_1, v_2, \dots, v_m\}$ represents the set of all accounts in the financial system, where each node v_i corresponds to an account A_i with associated financial, categorical, and regulatory attributes. The directed edge set $\mathcal{E}^{(t)} \subseteq \mathcal{V} \times \mathcal{V}$ captures temporal financial flows such that an edge $(v_i, v_j) \in \mathcal{E}^{(t)}$ indicates a transaction from account A_i to account A_j at time t (As shown in Figure 6).

Each node v_i carries a feature vector $\mathcal{X}_i^{(t)} \in \mathbb{R}^d$, where d includes components such as beginning balance, account category encoding, policy compliance indicators, jurisdictional region code, and any exogenous inputs relevant to audit risk or transaction frequency. Similarly, edge features $\mathcal{F}_{ij}^{(t)} \in \mathbb{R}^k$ store the transaction metadata such as amount, currency type, transaction latency, settlement flags, and contractual terms.

The edge features are derived from the symbolic accounting transaction matrix $T^{(t)} \in \mathbb{R}^{m \times m}$. Each scalar $T_{ij}^{(t)}$ denotes a monetary transfer from account A_i to account A_j . We define a contextualized feature constructor $\phi : \mathbb{R} \times \Theta \rightarrow \mathbb{R}^k$, such that

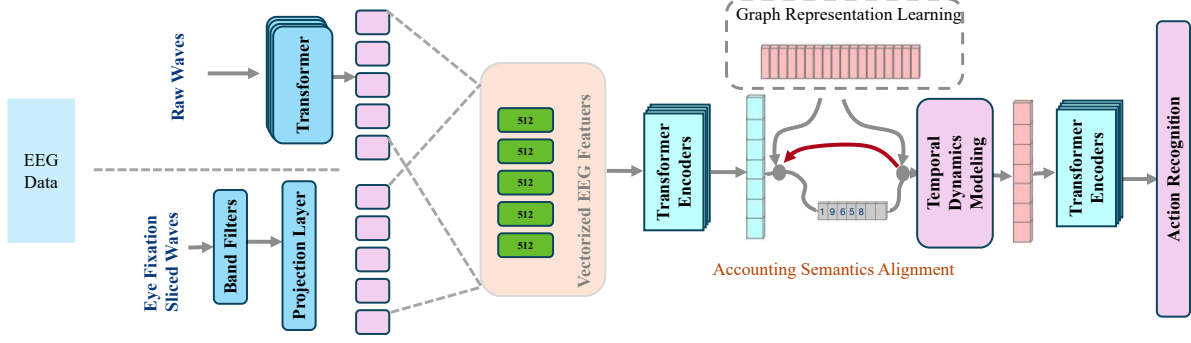


Figure 5. Illustration of Graph-Fused Accounting Network (GFA-Net). This diagram outlines the end-to-end process of transforming raw EEG data—comprising raw waves and eye fixation signals—into action recognition outputs. The pipeline includes preprocessing via band filters and projection layers, feature extraction using transformers, graph-based representation learning, temporal dynamics modeling, semantic alignment, and final recognition through additional transformer encoders.

$$\mathcal{F}_{ij}^{(t)} = \phi \left(T_{ij}^{(t)}, \theta_{ij}^{(t)} \right), \quad (6)$$

where $\theta_{ij}^{(t)}$ denotes auxiliary attributes including risk score, contract duration, and compliance tags. To enhance interpretability, we apply domain-based standardization on $\phi(\cdot)$ ensuring invariant scaling across time periods.

The initial node embedding matrix $\mathbf{H}^{(0)} \in \mathbb{R}^{m \times d}$ is initialized by concatenating raw node features and any policy regime embedding $\mathbf{e}_p^{(t)}$ active during time t . The graph encoding module uses a stack of edge-aware GNN layers, where the l -th layer updates each node embedding $\mathbf{h}_i^{(l)}$ via neighborhood aggregation and transformation.

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \psi \left(\mathbf{h}_j^{(l)}, \mathcal{F}_{ji}^{(t)} \right) + \mathbf{W}_1 \mathbf{h}_i^{(l)} \right), \quad (7)$$

where $\psi(\cdot)$ is a learned edge-conditioned message function and $\sigma(\cdot)$ is a nonlinear activation function such as ReLU or GELU. $\mathcal{N}(i)$ is the set of all incoming neighbors of node v_i , and $\mathbf{W}_1 \in \mathbb{R}^{d \times d'}$ is a learnable transformation matrix.

To model residual structure in node updates and improve gradient flow in deep layers, we augment the update rule with skip connections.

$$\mathbf{h}_i^{(l+1)} = \mathbf{h}_i^{(l)} + \sigma \left(\mathbf{W}_2 \cdot \sum_{j \in \mathcal{N}(i)} \psi \left(\mathbf{h}_j^{(l)}, \mathcal{F}_{ji}^{(t)} \right) \right), \quad (8)$$

where $\mathbf{W}_2$ is a projection matrix ensuring dimensional alignment across layers. Optionally, layer normalization or batch normalization can be inserted between summation and activation.

To incorporate global context into the graph structure, we include a graph-level summary vector computed via attention pooling.

$$\mathbf{g}^{(t)} = \sum_{i=1}^m \alpha_i^{(t)} \cdot \mathbf{h}_i^{(L)}, \quad \text{with } \alpha_i^{(t)} = \frac{\exp(\mathbf{a}^\top \mathbf{h}_i^{(L)})}{\sum_{j=1}^m \exp(\mathbf{a}^\top \mathbf{h}_j^{(L)})}, \quad (9)$$

where $\mathbf{a} \in \mathbb{R}^{d'}$ is a learnable attention vector. This enables downstream modules to condition on global fiscal context in a differentiable manner.

The output of the GNN stack is consolidated into the encoded node-state matrix.

$$\mathbf{H}^{(t)} = [\mathbf{h}_1^{(L)}, \mathbf{h}_2^{(L)}, \dots, \mathbf{h}_m^{(L)}]^\top \in \mathbb{R}^{m \times d'}, \quad (10)$$

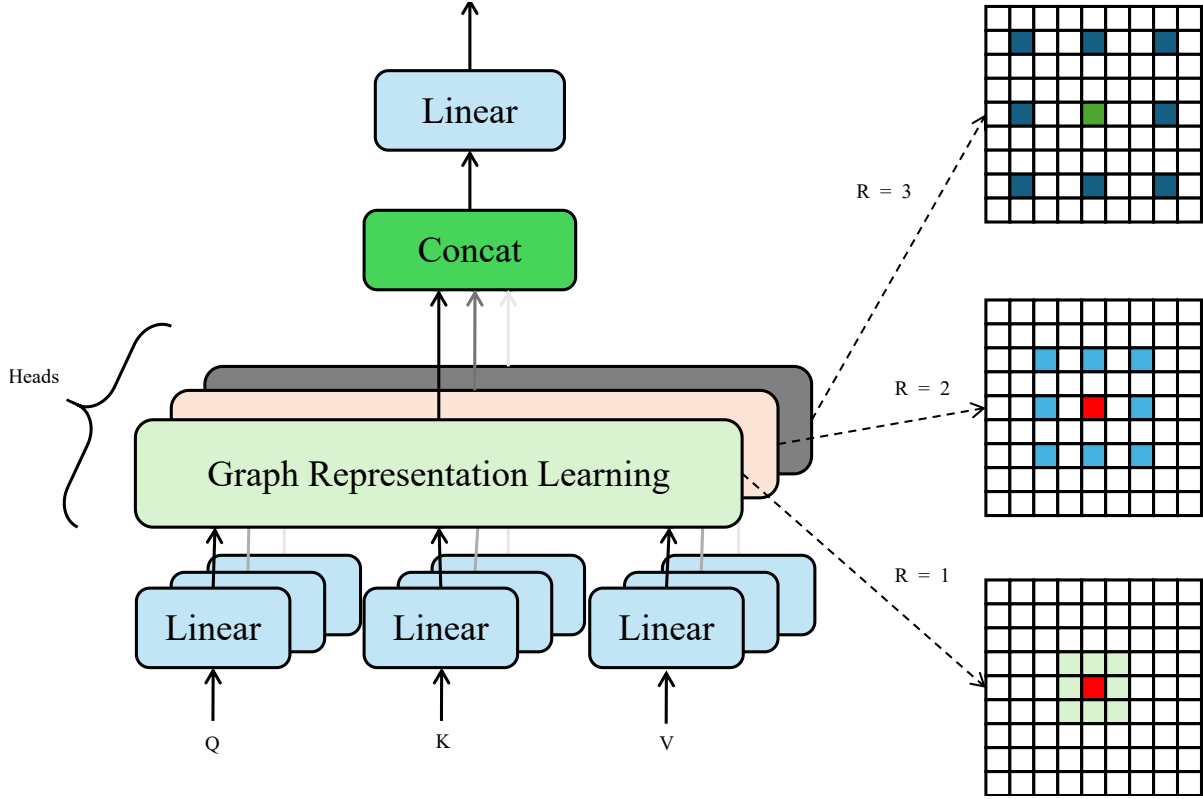


Figure 6. Illustration of Graph Representation Learning. This diagram visualizes how multi-head attention is integrated with graph representation learning. The process begins with linear transformations of query (Q), key (K), and value (V) inputs. These are passed through multiple attention heads, each incorporating graph-based structures at varying receptive field radii ($R = 1, 2, 3$). The outputs are concatenated and passed through a final linear layer, enhancing spatial awareness and contextual aggregation for improved representation learning.

which represents semantically-enriched, policy-aware financial account embeddings at time t , suitable for projection into report generation, compliance validation, and scenario simulation pipelines.

Accounting Semantics Alignment

To preserve the semantics of accounting laws within the learned representations, GFA-Net introduces a differentiable projection layer that enforces identity consistency and institutional compliance at both the balance sheet and income statement levels. Let $\widehat{\mathbf{B}}^{(t)} \in \mathbb{R}^m$ denote the predicted vector of account balances at time t . Each component $\widehat{B}_i^{(t)}$ corresponds to the learned financial representation of account A_i produced by the GNN encoder. The system must satisfy the fundamental balance sheet identity.

$$\sum_{A_i \in \text{Assets}} \widehat{B}_i^{(t)} = \sum_{A_j \in \text{Liabilities}} \widehat{B}_j^{(t)} + \sum_{A_k \in \text{Equity}} \widehat{B}_k^{(t)}. \quad (11)$$

To enforce this constraint in a differentiable form, we define a quadratic penalty loss using indicator vectors $\mathbf{C}_a$, $\mathbf{C}_l$, and $\mathbf{C}_e \in \{0, 1\}^m$ which select the components of $\widehat{\mathbf{B}}^{(t)}$ belonging to assets, liabilities, and equity classes, respectively.

$$\mathcal{L}_{\text{id}}^{(t)} = \left\| \mathbf{C}_a^\top \widehat{\mathbf{B}}^{(t)} - \left(\mathbf{C}_l^\top \widehat{\mathbf{B}}^{(t)} + \mathbf{C}_e^\top \widehat{\mathbf{B}}^{(t)} \right) \right\|_2^2. \quad (12)$$

In parallel, income verification is carried out by ensuring that the predicted net income aligns with the difference between revenue and expense components. Let $\mathcal{R}$ and $\mathcal{E}$ denote index sets of revenue and expense accounts. Then the predicted income is computed.

$$\widehat{\text{Net Income}}^{(t)} = \sum_{i \in \mathcal{R}} \widehat{B}_i^{(t)} - \sum_{j \in \mathcal{E}} \widehat{B}_j^{(t)}. \quad (13)$$

To incorporate this into the training objective, we define the income loss.

$$\mathcal{L}_{\text{inc}}^{(t)} = \left(\widetilde{\text{Net Income}}^{(t)} - \widehat{\text{Net Income}}^{(t)} \right)^2, \quad (14)$$

where $\widetilde{\text{Net Income}}^{(t)}$ is a ground-truth or externally estimated value derived from authoritative financial records or audit-certified labels. Moreover, to ensure the smooth integration of these symbolic constraints into the end-to-end learning pipeline, we define a projected balance vector using an alignment operator $\Pi(\cdot)$.

$$\widetilde{\mathbf{B}}^{(t)} = \Pi(\widehat{\mathbf{B}}^{(t)}) = \arg \min_{\mathbf{b} \in \mathbb{R}^m} \left\| \mathbf{b} - \widehat{\mathbf{B}}^{(t)} \right\|_2^2 \quad \text{s.t.} \quad \mathbf{C}_a^\top \mathbf{b} = \mathbf{C}_l^\top \mathbf{b} + \mathbf{C}_e^\top \mathbf{b}. \quad (15)$$

This projection ensures the network's output can be corrected post hoc without discarding learned statistical structure. A reporting fidelity loss is introduced to constrain deviations in key line-item summaries. Let $W \in \mathbb{R}^{d \times m}$ be the reporting aggregation matrix, and $\mathbf{r}_{\text{true}}^{(t)}$ the ground-truth report vector. The constraint reads.

$$\mathcal{L}_{\text{report}}^{(t)} = \left\| W \cdot \widehat{\mathbf{B}}^{(t)} - \mathbf{r}_{\text{true}}^{(t)} \right\|_2^2. \quad (16)$$

These semantic alignment components are jointly optimized with predictive and temporal objectives, enabling the model to remain faithful to financial structure while adapting to diverse operational contexts.

Temporal Dynamics Modeling

To capture the progression of financial states over time, GFA-Net employs a recurrent propagation module that explicitly models temporal dependencies between consecutive accounting periods. Let $\mathbf{H}^{(t)} \in \mathbb{R}^{m \times d'}$ denote the encoded financial graph representation at time t . These graph-level embeddings are summarized and passed into a recurrent neural unit to maintain an evolving state vector $\mathbf{z}^{(t)} \in \mathbb{R}^{d''}$, which captures long-term contextual dependencies across reporting periods. The update rule is governed by a gated recurrent unit (GRU) defined.

$$\mathbf{z}^{(t+1)} = \text{GRU}(\mathbf{H}^{(t)}, \mathbf{z}^{(t)}), \quad (17)$$

where $\mathbf{z}^{(t)}$ represents the accumulated knowledge prior to time t . The GRU learns temporal interactions between financial indicators and structural changes in the accounting graph. To encourage the model to generate temporally coherent predictions, we define a temporal smoothness loss by projecting the latent memory to the next-step prediction of account balances.

$$\mathcal{L}_{\text{temporal}} = \sum_{t=1}^{n-1} \left\| \widehat{\mathbf{B}}^{(t+1)} - f_{\text{proj}}(\mathbf{z}^{(t)}) \right\|_2^2, \quad (18)$$

where $f_{\text{proj}} : \mathbb{R}^{d''} \rightarrow \mathbb{R}^m$ is a neural projection function mapping recurrent states to predicted balances. This temporal constraint enables the model to learn plausible transitions between periods, consistent with real-world accounting flows.

The anomaly detection mechanism in GFA-Net evaluates the output of each account node for irregular patterns by applying a binary classifier η over the final graph layer embeddings $\mathbf{h}_i^{(L)}$. For each account i , the binary cross-entropy loss is used to supervise the prediction against ground truth anomaly labels $y_i^{(t)}$.

$$\mathcal{L}_{\text{anomaly}} = \sum_{i=1}^m \text{BCE}(y_i^{(t)}, \eta(\mathbf{h}_i^{(L)})), \quad (19)$$

where $\text{BCE}(\cdot)$ denotes the binary cross-entropy function and $\eta : \mathbb{R}^{d'} \rightarrow [0, 1]$ is a learned sigmoid classifier. A compliance score is computed for each period t to reflect whether the temporal state aligns with regulatory feasibility. This score is modeled via a linear transformation followed by a sigmoid activation.

$$\mathcal{L}_{\text{compliance}} = -\log \left(\sigma \left(\mathbf{w}_c^\top \mathbf{z}^{(t)} + b_c \right) \right), \quad (20)$$

where $\mathbf{w}_c \in \mathbb{R}^{d''}$ and $b_c \in \mathbb{R}$ are trainable weights and $\sigma(\cdot)$ denotes the sigmoid function.

5.4 Policy-Consistent Scenario Flow Strategy (PCS-Flow)

To complement the modeling capability of GFA-Net, we introduce PCS-Flow (Policy-Consistent Scenario Flow Strategy), a regulatory-aware, scenario-adaptive strategic framework that governs the behavior of financial state evolution under variable accounting regimes. PCS-Flow is designed to align neural propagation and decision reasoning with heterogeneous accounting standards, fiscal regulations, and audit interpretations across jurisdictions and temporal regimes. The strategy operates through a series of differentiable transformations and constraint regularizers that encode legal consistency, policy coherence, and temporal resilience (As shown in Figure 7).

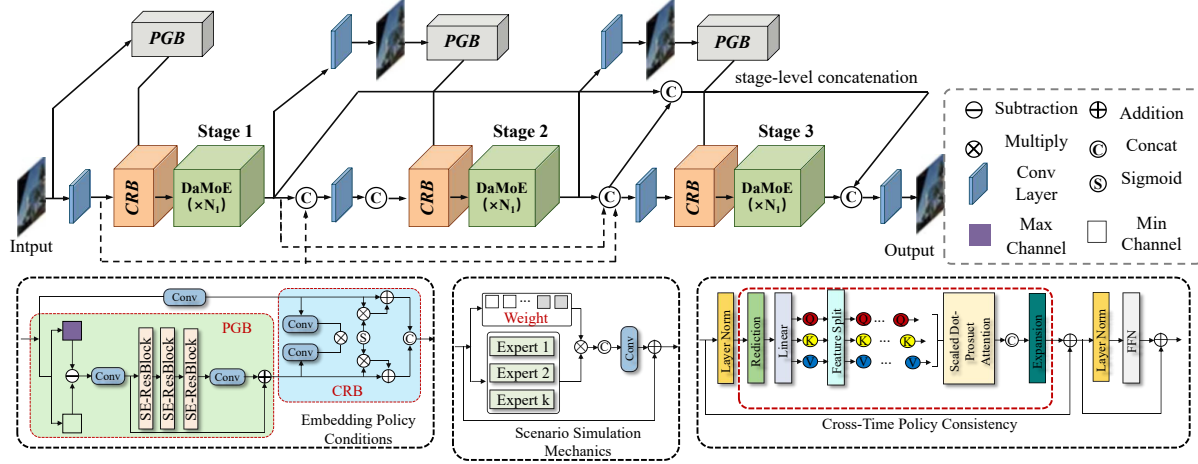


Figure 7. Illustration of the Policy-Consistent Scenario Flow Strategy (PCS-Flow). This diagram presents a three-stage architecture combining Conditional Residual Blocks (CRB) and Dynamic Mixture-of-Experts (DaMoE) modules, with stage-wise concatenation. Each stage refines the input through Policy-Guided Blocks (PGB) and CRB-enhanced feature extraction. The design includes an embedding policy mechanism, scenario simulation via expert gating, and a cross-time policy consistency module leveraging attention mechanisms and feature splitting for temporal alignment and robust output generation.

Embedding Policy Conditions

Let $\mathcal{P} = \{p_1, p_2, \dots, p_s\}$ denote a finite set of discrete accounting policy regimes, such as IFRS, US GAAP, or jurisdiction-specific tax frameworks. Each policy regime p_k is associated with a set of semantic and structural interpretations regarding recognition, measurement, and disclosure rules. To parameterize these differences, we define a policy embedding matrix $\mathbf{E}_{\mathcal{P}} \in \mathbb{R}^{s \times d_p}$, where each row $\mathbf{E}_{\mathcal{P}}[k, :]$ corresponds to the latent representation of policy p_k . These embeddings are learned end-to-end and encode differential assumptions, such as treatment of leasing, goodwill impairment, or revenue recognition timing (As shown in Figure 8).

Each accounting period t is labeled with an active policy tag $p^{(t)} \in \mathcal{P}$, indicating the applicable reporting framework. The corresponding embedding vector is extracted.

$$\mathbf{e}_p^{(t)} = \mathbf{E}_{\mathcal{P}}[p^{(t)}], \quad (21)$$

which is used to inform graph construction and neural encoding for that period. The policy vector $\mathbf{e}_p^{(t)}$ is concatenated to both the node-level account features and edge-level transaction features to contextualize them under the governing regime.

$$\mathcal{X}_i^{(t)} \leftarrow [\mathcal{X}_i^{(t)} \parallel \mathbf{e}_p^{(t)}], \quad \mathcal{F}_{ij}^{(t)} \leftarrow [\mathcal{F}_{ij}^{(t)} \parallel \mathbf{e}_p^{(t)}], \quad (22)$$

for all $i, j \in \{1, \dots, m\}$. This ensures that policy-conditional encoding affects all message passing operations and node updates in GFA-Net.

To increase the expressiveness of $\mathbf{e}_p^{(t)}$ beyond static embeddings, we introduce a trainable transformation function $g_{\text{policy}} : \mathbb{R}^{d_p} \rightarrow \mathbb{R}^{d'}$ that maps raw policy vectors into a space aligned with GNN feature representations.

$$\tilde{\mathbf{e}}_p^{(t)} = g_{\text{policy}}(\mathbf{e}_p^{(t)}) = \text{MLP}(\mathbf{e}_p^{(t)}), \quad (23)$$

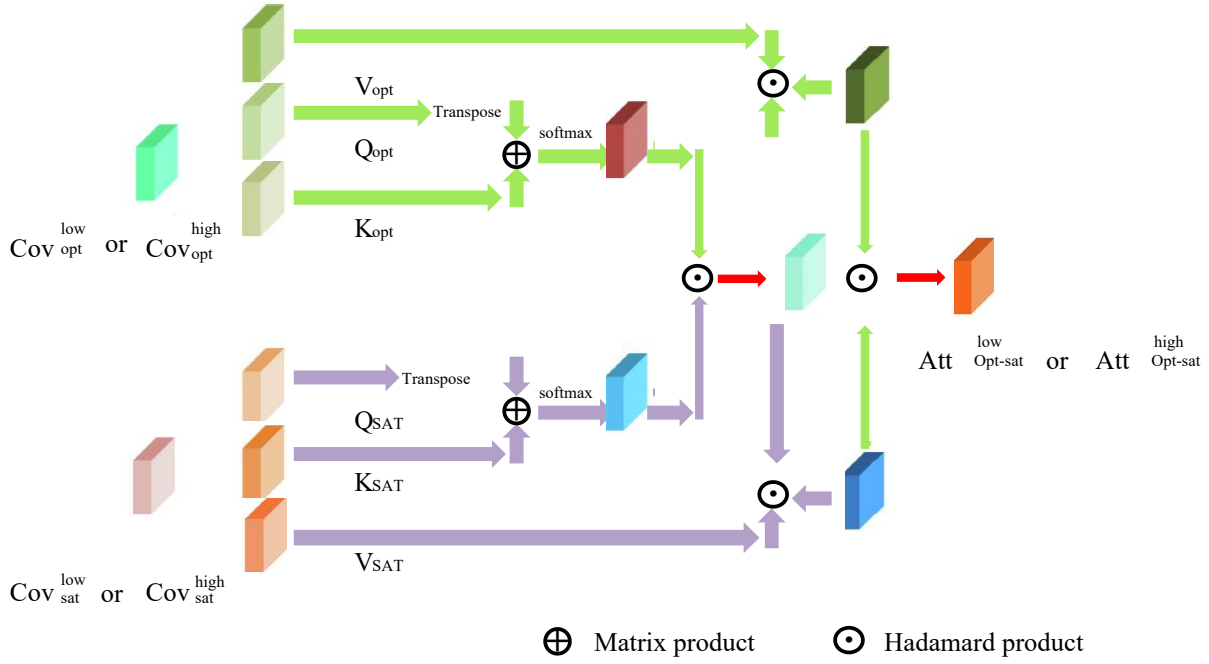


Figure 8. Illustration of Embedding Policy Conditions. This figure shows a comparison between two parallel attention computation pathways including one under optimal covariance conditions (Cov_{opt}) and another under saturation conditions (Cov_{sat}). Both paths utilize query (Q), key (K), and value (V) vectors—denoted as Q_{opt} , K_{opt} , V_{opt} and Q_{SAT} , K_{SAT} , V_{SAT} , respectively. The attention outputs are derived via matrix products followed by softmax and Hadamard products, leading to either low or high attention outputs ($Att_{Opt-sat}$). This dual-path structure emphasizes the role of attention state variation in modeling adaptive responses.

where MLP denotes a multi-layer perceptron with non-linear activations. The transformed vector $\tilde{\mathbf{e}}_p^{(t)}$ can then be shared across both node and edge updates, as well as temporal modules such as GRU or scenario simulators.

To quantify the influence of policy variability across time, we define a policy transition loss between adjacent periods. Let $\mathbf{e}_p^{(t)}$ and $\mathbf{e}_p^{(t+1)}$ denote embeddings for two consecutive periods. The smoothness constraint is defined.

$$\mathcal{L}_{\text{policy-shift}}^{(t)} = \left\| \mathbf{e}_p^{(t+1)} - \mathbf{e}_p^{(t)} \right\|_2^2, \quad (24)$$

penalizing abrupt regime transitions unless externally justified. We define a regime classification head to ensure that embeddings encode distinguishable semantics. Let $h_{\text{cls}} : \mathbb{R}^{d_p} \rightarrow \mathbb{R}^s$ be a softmax classifier, then the auxiliary classification loss is given.

$$\mathcal{L}_{\text{regime-cls}} = -\log \left(\frac{\exp(h_{\text{cls}}(\mathbf{e}_p^{(t)})[p^{(t)}])}{\sum_{k=1}^s \exp(h_{\text{cls}}(\mathbf{e}_p^{(t)})[k])} \right). \quad (25)$$

This supervision encourages the policy vectors to remain identifiable in latent space, enabling better generalization across unseen jurisdictions. These mechanisms collectively embed structural regulatory assumptions directly into the computational pipeline.

Scenario Simulation Mechanics

We define a set of hypothetical scenarios $\mathcal{S} = \{s_1, s_2, \dots, s_r\}$, each corresponding to a macro-financial perturbation or policy-sensitive event, such as inflation surges, regulatory amendments, or market liquidity shocks. Each scenario s_j is formalized as a transformation operator $\mathcal{T}_{s_j}$ that perturbs the input node features of the accounting graph, modeling how external shocks reconfigure the financial landscape.

For each account i at time t , the perturbed feature under scenario s_j is computed.

$$\mathcal{X}_{i,s_j}^{(t)} = \mathcal{T}_{s_j}(\mathcal{X}_i^{(t)}) = \mathcal{X}_i^{(t)} + \Delta_{i,s_j}^{(t)}, \quad (26)$$

where $\Delta_{i,s_j}^{(t)}$ encodes scenario-specific deviations. These perturbed node features define a new scenario-specific graph $G_{s_j}^{(t)}$. The GFA-Net encoder is applied over the transformed graph to generate scenario-specific hidden representations.

$$\mathbf{H}_{s_j}^{(t)} = \text{GFA-Net}(G^{(t)}[\mathcal{X}_{s_j}^{(t)}]), \quad (27)$$

yielding updated account embeddings that reflect the simulated environment. These are decoded into scenario-projected balances $\widehat{\mathbf{B}}_{s_j}^{(t)}$, which are then aggregated into structured financial reports.

$$\mathbf{r}_{s_j}^{(t)} = W \cdot \widehat{\mathbf{B}}_{s_j}^{(t)}, \quad (28)$$

with W denoting the reporting aggregation matrix. To maintain semantic coherence across accounting regimes, PCS-Flow introduces a policy-consistency constraint. Given a base policy p_b and a target policy p_a .

$$\mathcal{L}_{\text{policy}}^{(t)} = \left\| \mathbf{r}_{p_b}^{(t)} - \Pi_{p_a \leftarrow p_b}(\mathbf{r}_{p_b}^{(t)}) \right\|_2^2, \quad (29)$$

where $\Pi_{p_a \leftarrow p_b}(\cdot)$ is a learned operator translating reporting semantics from p_b to p_a . To evaluate scenario desirability under uncertainty, PCS-Flow ranks alternatives using a regulatory-compliance-weighted utility score.

$$\rho_j^{(t)} = \alpha \cdot \gamma_j^{(t)} + (1 - \alpha) \cdot f_{\text{util}}(\mathbf{r}_{s_j}^{(t)}), \quad (30)$$

where $\gamma_j^{(t)}$ is the compliance confidence score, $f_{\text{util}}(\cdot)$ is a task-specific utility function, and $\alpha \in [0, 1]$ balances feasibility and financial performance.

Cross-Time Policy Consistency

We apply a smoothness prior to ensure that reported metrics evolve consistently across time within the same policy. For each t .

$$\mathcal{L}_{\text{temporal-consistency}}^{(t)} = \left\| \mathbf{r}^{(t+1)} - \mathbf{r}^{(t)} \right\|_{\Sigma_p^{-1}}^2, \quad (31)$$

where Σ_p is a learned covariance structure specific to policy $p^{(t)}$, capturing expected volatility between consecutive periods. For decision support, PCS-Flow provides scenario ranking under regulatory feasibility and financial desirability. Each scenario s_j receives a compliance score γ_j and a utility score v_j .

$$\gamma_j = \sigma(\mathbf{w}_\gamma^\top \mathbf{H}_{s_j}^{(t)} + b_\gamma), \quad v_j = f_{\text{utility}}(\mathbf{r}_{s_j}^{(t)}), \quad (32)$$

with final ranking score.

$$\rho_j = \alpha \cdot \gamma_j + (1 - \alpha) \cdot v_j, \quad (33)$$

where $\alpha \in [0, 1]$ balances legal and performance objectives.

The integrated PCS-Flow objective is constructed.

$$\mathcal{L}_{\text{PCS}} = \sum_{t=1}^n \left(\lambda_5 \mathcal{L}_{\text{policy}}^{(t)} + \lambda_6 \mathcal{L}_{\text{temporal-consistency}}^{(t)} \right) + \lambda_7 \sum_{j=1}^r (1 - \gamma_j)^2, \quad (34)$$

subject to compliance constraints.

$$\Gamma(\mathbf{r}_{s_j}^{(t)}) = 1, \quad \forall j, t. \quad (35)$$

PCS-Flow ensures that financial predictions remain robust under changing external and internal conditions, while maintaining interpretability and legal coherence.

Conflict of Interest Statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Author Contributions

Conceptualization, TQ; methodology, TQ; software, TQ; validation, TQ; formal analysis, TQ; investigation, TQ; data curation, TQ; writing—original draft preparation, TQ; writing—review and editing, TQ; visualization, TQ; supervision, TQ; funding acquisition, TQ. All authors have read and agreed to the published version of the manuscript.

Funding

Details of all funding sources should be provided, including grant numbers if applicable. Please ensure to add all necessary funding information, as after publication this is no longer possible.

Acknowledgments

This is a short text to acknowledge the contributions of specific colleagues, institutions, or agencies that aided the efforts of the authors.

Data Availability Statement

The datasets generated and/or analysed during the current study are available in the ShadowBank, <https://github.com/aeon38116/ShadowBank.git>

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